How ChatVault is built

A single HTML file that runs entirely in the browser, backed by Python tools for batch work and an AI layer for summaries and agentic tasks. No server and no installation.

DATA FLOW

INPUT
WhatsApp export
.zip with media or .txt, Android or iPhone

→

PROCESS IN BROWSER
Unzip, hash, parse, link
Merge exports, analyse, apply roles and redaction

→

STORE
Encrypted archive
Browser database, folder, online drive

→

OUTPUT
Evidence files
PDF, Excel, Word, ZIP, all hashed

TECHNOLOGY STACK

Interface	**HTML5, CSS3, JavaScript (ES2022)** · single-file web app, WhatsApp-style layout, light and dark themes, six colour schemes, phone to desktop
Processing	**JSZip** to read exports · own parser for Android and iPhone formats · merge-and-compare engine for repeat exports
Database (DBMS)	**IndexedDB** (the browser's built-in database) · **localStorage** for accounts and audit log · **File System Access API** for a local or Google Drive / OneDrive folder
Security	**Web Crypto API** · SHA-256 fingerprints · AES-256-GCM encryption · PBKDF2-SHA-256 password hashing (150,000 iterations)
Documents	**pdfmake** with Noto Sans Gujarati and Devanagari · **SheetJS** for Excel · **pdf.js** viewer · Office Open XML for the Word certificate
Python	**chatvault.py** command-line archiver (ReportLab, openpyxl) · **filing_agent.py** verifies and files archives · sample generator (Pillow, FFmpeg)
Artificial intelligence	**Claude** summary, action items, timeline and amounts, each linked to its message · rule-based insights offline
Agentic browser	Supervised task prompts: upload archives to Google Drive with restricted sharing, check GSTINs on the GST portal

DATA MODEL

`chats` · encrypted details, tags, notes, custody log

`blobs` · encrypted original exports, keyed by SHA-256

`users` · `audit` · salted hashes, wrapped keys, actions

`folder` · vault.json and encrypted .cvault files

SECURITY AND PRIVACY

- ✓ No server: processing stays on the device
- ✓ One vault key, wrapped with each user's password
- ✓ Fingerprints re-checked every time a chat opens
- ✓ Roles, lockout, auto-lock, privacy screen, redaction

Tested end to end
Automated browser tests (Playwright, headless Chrome): import, merge, exports, roles, encryption, folder sync, phone layout

Submitted on GitHub
`ChatVault.html` (offline) · desktop shortcut with icon · one-page PDFs · source, Python tools, agent, samples